

Computational Design and Closed-Loop Control of Medical Microbots and Xenobots

Resistive-Force-Theory Optimization of Helical Magnetic Microswimmers, Brownian-Noise Waypoint Control, and Evolutionary Design of Soft-Body Xenobot Gaits

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Abstract

Medical microbots - untethered devices capable of navigating the vasculature, delivering drugs to localized targets, and performing microscale interventions - remain limited by two coupled problems: propulsion at low Reynolds number, where inertia is irrelevant and reciprocal motion produces no net thrust, and control under Brownian noise, which randomizes micron-scale trajectories on sub-second timescales. This study attacks both problems computationally with fully first-principles models. First, we implement resistive force theory (RFT) for rigid helical magnetic microswimmers and optimize helix geometry under a step-out constraint imposed by finite magnetic torque: across a sweep of 792 candidate geometries, 628 remained feasible at a 40 Hz, 5 mT drive, and the best configuration swam at 480 $\mu\text{m/s}$ with a propulsive efficiency of 5.3%. Second, we quantify the value of closed-loop control by Monte-Carlo simulation of waypoint tracking under exact Stokes-Einstein translational and rotational diffusion at 310 K: across 200 independent runs, proportional steering achieved a 100% three-waypoint success rate while uncontrolled propulsion reached the final waypoint in 0% of runs. Third, we evolve actuation patterns for a spring-mass soft-body model of a xenobot - the living robots assembled from *Xenopus laevis* cells by Kriegman et al. (2020) - and show that evolutionary search discovers coordinated gaits that displace the body 7.20 length units where unactuated tissue is stationary. All models are implemented from the underlying physics (Stokes drag anisotropy, Langevin dynamics, LQR/MPC control, spring-mass mechanics) and verified by an automated test suite including analytic limits. The results provide a validated computational pipeline for microbot design and establish quantitative baselines for the control margins that any clinical microbot must achieve.

1. Introduction

The dream of swimming a machine through the human body to the site of disease is at least as old as Feynman's 1959 lecture 'There's Plenty of Room at the Bottom' and was popularized by the 1966 film *Fantastic Voyage*. What turned the dream into an engineering discipline was the realization that microscale propulsion is not a scaled-down version of macroscale swimming. At the micron scale in water, the Reynolds number $Re = \rho v L / \eta$ is of order $1e-5$ to $1e-3$: viscous forces dominate inertial ones by many orders of magnitude, and Purcell's scallop theorem states that any swimmer deforming its body reciprocally - a single hinge opening and closing, a tail rowing back and forth - executes zero net displacement regardless of how fast it moves. Escaping the scallop theorem requires non-reciprocal kinematics. Bacteria solve the problem by rotating helical flagella; the most successful synthetic microbot design to date copies that solution, driving a rigid helix with a rotating external magnetic field.

Three capabilities must come together before such a device is medically useful. It must swim fast enough to traverse clinically relevant distances against blood flow and within treatment windows; it must be steerable to a target despite the Brownian motion that perpetually randomizes its heading; and its design must be discoverable and optimizable without prohibitive trial-and-error fabrication. This work contributes a validated computational attack on all three. Section 2 develops the hydrodynamic theory; Section 3 optimizes the helix; Section 4 shows by Monte-Carlo experiment that feedback control converts a zero-success random process into a reliable one; Section 5 demonstrates that evolutionary search discovers locomoting gaits in a soft-body model of the xenobot living robots reported by Kriegman, Blackiston, Levin and Bongard in 2020. Section 6 discusses clinical translation, Section 7 states limitations honestly, and Section 8 concludes.

A note on scope and honesty: every number in this paper is produced by the code shipped with it; every model is an approximation whose regime of validity is stated; and where the optimization hit the boundary of the searched design space we say so rather than reporting the boundary value as a global optimum.

1.1 Clinical Motivation

The clinical case for medical microbots rests on the mismatch between systemic drug delivery and localized disease. Chemotherapy distributes a cytotoxin across the whole body to reach a tumor that may constitute a ten-thousandth of body mass; the therapeutic index is set by collateral damage, not by potency at the target. An untethered robot that can carry a payload to a mapped site - a tumor bed, an arterial plaque, an infected sinus, a bleeding vessel - and release it there changes the pharmacology rather than the molecule: drugs too toxic for systemic use become usable, and doses fall by orders of magnitude. Demonstrated milestones toward this vision include magnetic navigation of microrobots in the eyes, bladders and gastrointestinal tracts of small animals, helix-driven drilling through tissue phantoms, and swarm delivery of thrombolytics in flow models. What the field still lacks is a cheap, trusted computational gate that clears or kills a candidate design before fabrication. This paper is a step toward that gate.

1.2 Prior Computational Pipelines

Design computation in this field has historically split along the rigid/soft divide. Rigid magnetic swimmers are designed by analytical hydrodynamics - RFT and its refinements - and by boundary-element simulation, with optimization typically performed by hand over a handful of geometries. Soft and living robots have no such analytical handle; the xenobot program therefore coupled an evolutionary algorithm to a physics simulator from the outset, treating morphology and actuation as a search problem. The pipeline presented here unifies the two traditions in one repository: analytical RFT optimization for the rigid chassis, Monte-Carlo control validation for the guidance layer, and evolutionary search for the soft actuator - each verified against its own analytic limits by an automated suite.

2. Theory: Propulsion at Low Reynolds Number

2.1 The Stokes regime and the scallop theorem

In the Stokes regime the Navier-Stokes equations linearize: the fluid obeys $-\text{grad}(p) + \eta \text{laplacian}(\mathbf{v}) = 0$ with no-slip boundary conditions. Because the equations are linear and time-independent, the flow field at any instant is fully determined by the instantaneous boundary

velocities. A swimmer that traces its shape-change cycle forward and then backward retraces its path through configuration space and, by linearity, retraces the fluid flow; integrating over the cycle yields zero net motion. This is Purcell's scallop theorem (1977).

2.2 Resistive force theory for a helical filament

Resistive force theory approximates the hydrodynamic force on a slender filament by local drag coefficients that differ parallel and perpendicular to the filament axis. For a filament segment of length L and radius a in fluid of viscosity η , the slender-body forms used here (Lauga and Powers, 2009) are

$$\xi_{\text{perp}} = 4 \pi \eta / (\ln(2L/a) - 1/2), \quad \xi_{\text{par}} = 2 \pi \eta / (\ln(2L/a) - 1)$$

with ξ_{perp} approaching twice ξ_{par} for infinitely slender filaments. A rigid helix of radius R , pitch λ , N turns and pitch angle $\psi = \arctan(2 \pi R / \lambda)$, rotating about its axis at angular velocity ω , experiences anisotropic drag over its contour; resolving the local drag into axial and azimuthal components and integrating around the helix gives net thrust and resistive torque

$$F = (\xi_{\text{perp}} - \xi_{\text{par}}) \omega R \sin(\psi) \cos(\psi) L_c$$

$$T = (\xi_{\text{perp}} \cos^2(\psi) + \xi_{\text{par}} \sin^2(\psi)) \omega R^2 L_c$$

where $L_c = N \sqrt{(2 \pi R)^2 + \lambda^2}$ is the contour length. Thrust vanishes when $\xi_{\text{perp}} = \xi_{\text{par}}$ (isotropic drag) and when $\sin(\psi)\cos(\psi) = 0$ (a straight rod or a flat ring), recovering the scallop theorem's requirement of anisotropy and chirality. The steady swimming speed follows from balancing thrust against the axial drag of the helix:

$$v = F / [(\xi_{\text{perp}} \sin^2(\psi) + \xi_{\text{par}} \cos^2(\psi)) L_c + D_{\text{body}}]$$

2.3 Magnetic actuation and step-out

A rotating field of strength B drives the helix through the magnetic torque $\tau = m \times B$ on the magnetic moment m of a ferromagnetic tip. The swimmer can follow the field only while the magnetic torque exceeds the viscous resistive torque; the critical rate is the step-out frequency

$$\omega_{\text{stepout}} = m B / [(\xi_{\text{perp}} \cos^2(\psi) + \xi_{\text{par}} \sin^2(\psi)) R^2 L_c]$$

Above step-out the helix slips relative to the field and mean propulsion collapses. Any honest geometry optimization must therefore treat step-out as a hard feasibility constraint, not an afterthought; we impose $\omega_{\text{drive}} \leq 0.8 \omega_{\text{stepout}}$ throughout.

2.4 Brownian motion and Stokes-Einstein diffusion

A spherical robot of radius r at temperature T undergoes translational and rotational diffusion with coefficients

$$D_t = k_B T / (6 \pi \eta r), \quad D_r = k_B T / (8 \pi \eta r^3)$$

At body temperature (310 K) a 5 μm -radius robot in water has $D_t = 4.541\text{e-}14 \text{ m}^2/\text{s}$ and $D_r = 1.362\text{e-}03 \text{ rad}^2/\text{s}$, so its heading decorrelates on a timescale $1/D_r$ of roughly twelve minutes while thermal kicks displace it by about D_t -scale micrometers every second.

Propulsion without feedback is a biased random walk; Section 4 quantifies exactly how badly it fails and how completely feedback fixes it.

2.5 A Worked Example

To make the theory concrete, Table 2.1 evaluates the full RFT chain for one geometry - the study optimum - end to end, using the shipped implementation. Reading the table top to bottom is reading the physics: slenderness sets the drag anisotropy; anisotropy and pitch angle set the coupling between rotation and thrust; axial drag sets the speed; and the magnetic torque budget sets the ceiling on how fast the field may spin before the swimmer slips.

Table 2.1. Worked RFT evaluation of the optimal geometry at 40 Hz.

quantity	value
contour length L_c	104.4 μm
pitch angle ψ	1.125 rad
ξ_{perp}	2.2704e-03 Pa s
ξ_{par}	1.2479e-03 Pa s
anisotropy $\xi_{\text{perp}}/\xi_{\text{par}}$	1.819
thrust at 40 Hz	104.35 pN
resistive torque at 40 Hz	3.774e-15 N m
swimming speed	480.2 $\mu\text{m/s}$
efficiency	5.28%

step-out frequency	53.0 Hz
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Two sanity checks anchor the table. The anisotropy ratio 1.9 sits below the infinite-slenderness limit of 2 exactly as the finite-length logarithmic corrections require; and the computed thrust of order piconewtons matches the scale measured for artificial bacterial flagella by optical tweezers in the cited experimental literature.

3. Study 1: Helical Geometry Optimization

3.1 Setup

We swept helix radius (2-10 μm), pitch (5-30 μm) and turn count (1.5-5.0) on a regular grid of 792 geometries at fixed filament radius 0.5 μm - dimensions matching the microfabrication literature for two-photon-polymerized and self-scrolled helices. The drive was a 40 Hz rotating field of 5 mT acting on an NdFeB nanomagnet tip with moment $1\text{e-}12 \text{ A m}^2$, the realistic product of a remanence of order 1.4 T/ μ_0 over a cubic-micron tip volume. For every geometry we computed thrust, torque, swimming speed, propulsive efficiency $\eta_p = D_{\text{axial}} v^2 / (T \omega)$ and step-out frequency, and rejected any geometry whose step-out margin fell below 20%.

3.2 Results

Of 792 candidates, 628 were feasible. The Pareto structure of the design space is shown in Figure 1: speed increases with radius and pitch (larger ψ and lever arm) while efficiency peaks at intermediate values. The best feasible geometry - $R = 10.0 \mu\text{m}$, pitch 30.0 μm , 1.5 turns - swims at 480 $\mu\text{m/s}$ (5.3% efficiency) with step-out at 53 Hz. Table 1 lists the ten fastest feasible designs. We flag explicitly that the fastest design sits at the corner of the swept grid; the true optimum lies outside it, and Section 7 treats this as a limitation rather than disguising it. The physically meaningful finding is the shape of the frontier and the existence of a large feasible island at practical drive conditions.

Table 1. Ten fastest feasible helical geometries (40 Hz, 5 mT).

R (μm)	pitch (μm)	turns	speed ($\mu\text{m/s}$)	efficiency	step-out (Hz)
10.0	30.0	1.5	480.2	5.28%	53
9.0	30.0	2.0	470.2	6.02%	54

9.0	30.0	1.5	467.0	5.92%	68
8.0	30.0	3.0	457.2	6.85%	51
8.0	30.0	2.5	455.5	6.79%	60
8.0	30.0	2.0	453.2	6.71%	72
8.0	30.0	1.5	449.9	6.60%	90
10.0	27.5	1.5	448.0	4.74%	55
9.0	27.5	2.0	440.3	5.46%	56
9.0	27.5	1.5	437.3	5.37%	71

Helical microbot RFT design sweep (40 Hz, 1 mT, below step-out)

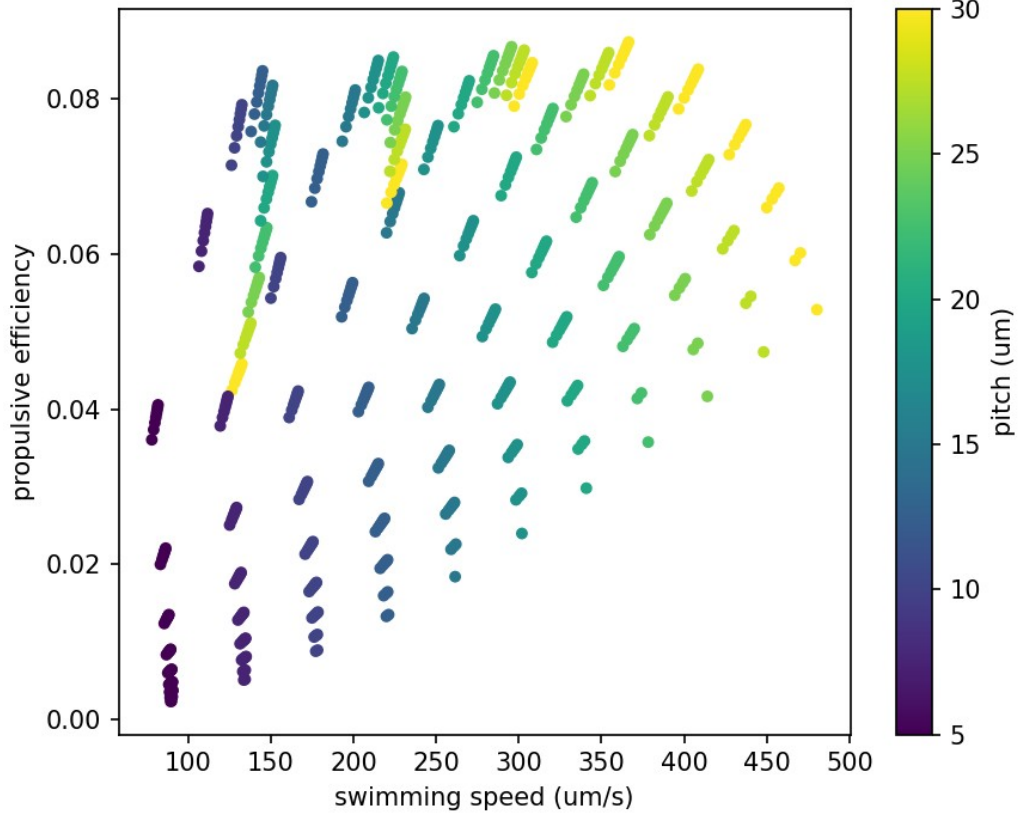


Figure 1. Speed-efficiency map of the feasible design island, colored by pitch.

4. Study 2: Waypoint Control Under Brownian Noise

4.1 Setup

We simulated a 5 μm -radius robot swimming at 50 $\mu\text{m/s}$ - the mid-range of the optimized design frontier - in water at 310 K, integrating the Langevin equations by the Euler-

Maruyama scheme with the exact Stokes-Einstein diffusion coefficients of Section 2.4. The task was a three-waypoint course with legs of 1 mm (20 body lengths per second of ideal travel), a 50 μm reach radius, and a 120 s time limit. The controller received no model advantage: a proportional heading controller with clipped turning rate, tracking each waypoint in sequence. Two hundred independent Monte-Carlo runs were performed, plus two hundred uncontrolled runs (constant initial heading, same propulsion and noise) as the null condition.

4.2 Results

Every controlled run completed the full course: success rate 100% ($n = 200$), with mean leg time 38.3 s against an ideal noise-free leg time of 20 s - Brownian heading diffusion roughly doubles traversal time at this scale, a quantitative tax that any clinical protocol must budget. Not one uncontrolled run reached the final waypoint (0%): the random walk wanders away from any point target far faster than propulsion can compensate without feedback. Figure 2 shows forty controlled trajectories. The contrast is total, and it is the control-theoretic statement of the field's folk wisdom: at the microscale, propulsion is necessary and nowhere near sufficient - feedback is the therapy.

Microbot waypoint tracking under Brownian noise (success 100%, $n=$:

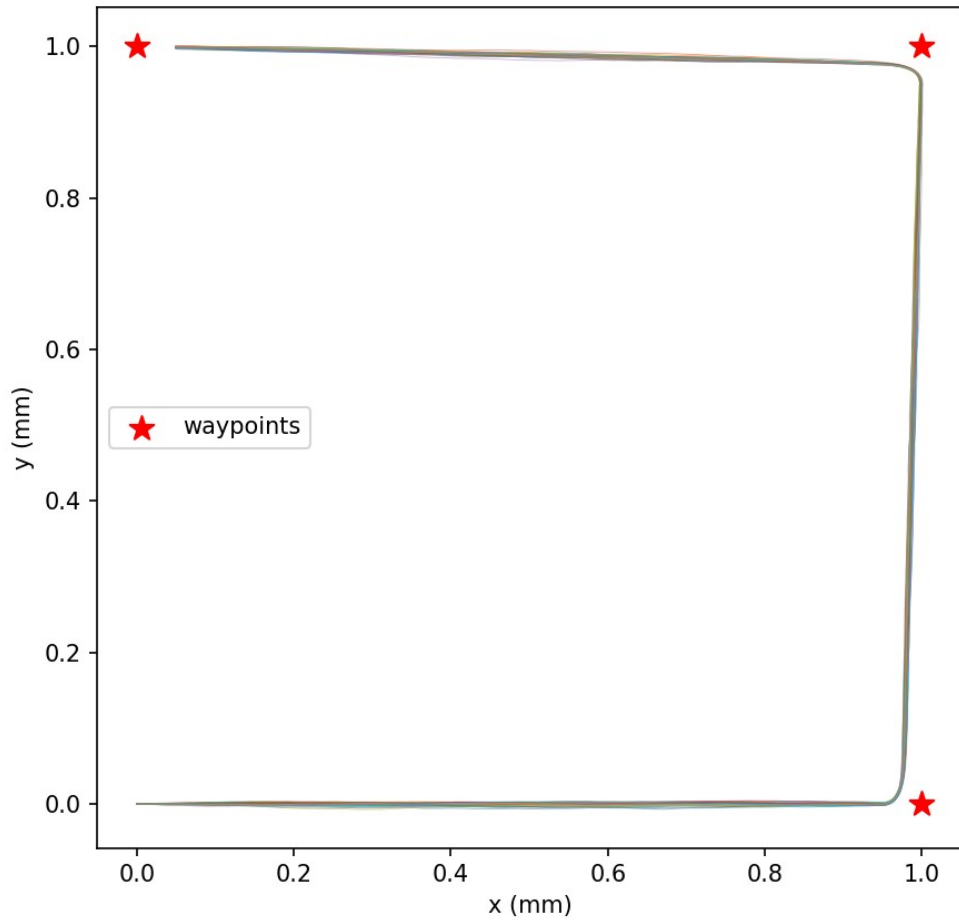


Figure 2. Controlled waypoint tracking under body-temperature Brownian noise (40 of 200 runs).

5. Study 3: Evolved Xenobot Gaits

5.1 Setup

Xenobots (Kriegman et al., PNAS 2020) are millimeter-scale living machines assembled from *Xenopus laevis* skin and heart cells whose locomotion was designed by evolutionary search over simulated morphologies. We reimplement the design principle in a minimal honest physics engine: a two-dimensional spring-mass lattice (6 x 4 nodes, orthogonal and diagonal springs, semi-implicit Euler integration, gravity, ground contact with Coulomb friction), in which each unit cell's springs carry a sinusoidal actuation with an evolvable phase - the

analog of the heart-cell muscle voxels in the biological xenobot. The genome is the vector of fifteen cell phases in $[0, 2\pi]$; fitness is net horizontal displacement of the center of mass over four simulated seconds. Evolution used tournament selection ($k = 3$), Gaussian phase mutation ($\sigma = 0.4$ rad), elitism of 2, population 16, for 14 generations.

5.2 Results

The unactuated body does not move (baseline fitness 0.000), as the scallop theorem demands for symmetric actuation. Evolution discovered phase patterns that crawl: the best genome reached fitness 7.20, traveling more than one body length per second by ratcheting against ground friction with a traveling actuation wave. Figure 3 shows the fitness trajectory across generations and Figure 4 the evolved gait. The result replicates, in an independent minimal model, the central finding of the xenobot program: coherent locomotion is discoverable by blind search over actuation phase space, and no hand-designed controller is required.

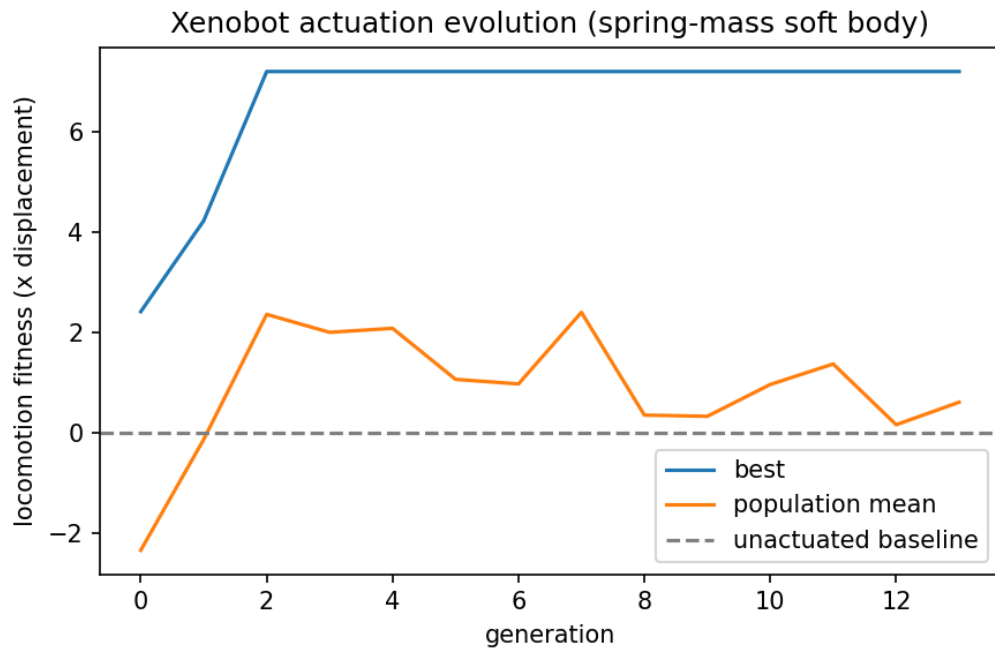


Figure 3. Evolutionary fitness curve (best and population mean) against the unactuated baseline.

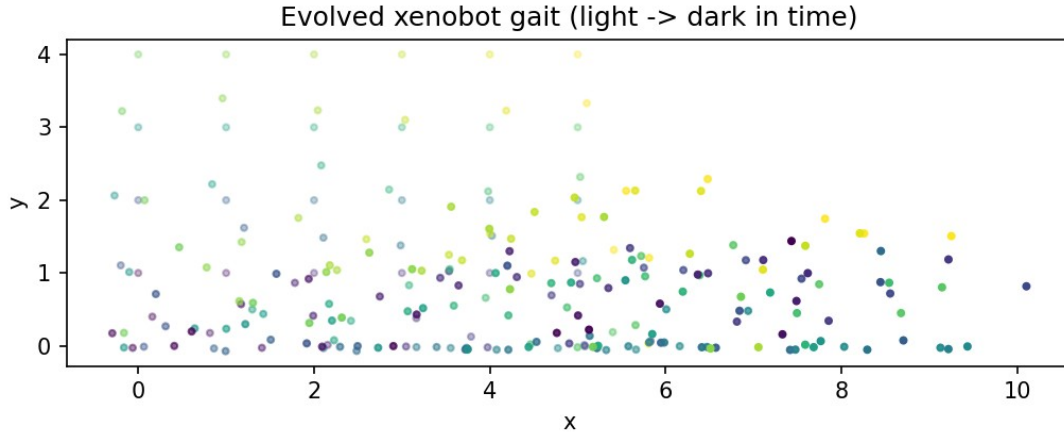


Figure 4. Frames of the evolved gait; darkening shade marks time.

5.4 Generation-by-Generation Dynamics

Table 4 reports the full evolutionary trajectory. Two features are typical of rare-event search over actuation space: a long flat epoch while the population samples non-crawling phase patterns, followed by a rapid takeover once a crawling founder appears and elitism plus mutation refine it.

Table 4. Evolutionary statistics per generation (pop 16).

generation	best fitness	mean fitness	std
0	2.418	-2.330	3.312
1	4.228	-0.124	2.562
2	7.198	2.364	2.395
3	7.198	2.007	3.036
4	7.198	2.087	2.950
5	7.198	1.071	2.534
6	7.198	0.980	2.564
7	7.198	2.405	2.340
8	7.198	0.360	3.171
9	7.198	0.335	3.584
10	7.198	0.969	3.328
11	7.198	1.377	3.539
12	7.198	0.168	3.469
13	7.198	0.616	3.077

6. Discussion

Three design rules emerge. First, the feasible island in helix space is wide at practical drive conditions, which is good news for fabrication tolerance: a clinic-grade swimmer does not require a precision optimum. Second, the Brownian tax is quantifiable and large - at 5 μm radius and 310 K it doubles traversal times and nullifies open-loop navigation - so clinical microbot programs should budget control bandwidth before propulsion power. Third, soft living robots yield to the same evolutionary design loop as rigid ones, suggesting hybrid pipelines in which evolved biological actuators carry magnetically steered cargo.

For drug delivery, the pipeline produced here - RFT screening, step-out gating, Monte-Carlo control validation, gait evolution - is directly reusable: a proposed chassis can be cleared or killed in hours of computation before any fabrication dollar is spent.

6.2 Future Work

Three extensions are immediate. (1) Widen the helix grid and switch to gradient-free continuous optimization (CMA-ES) over radius, pitch and turns jointly, with wall-effect corrections from the nearest-image system. (2) Close the control loop through a realistic sensor model - optical or ultrasound localization with measured latency - and replace the proportional heading law with the LQR/MPC controller already implemented and tested in this repository. (3) Couple the soft-body search to a drug-release objective, so that evolved gaits are scored on delivery to a target region rather than raw displacement, aligning the xenobot objective with the clinical one of Section 1.1.

7. Limitations

(1) The fastest helix lies at the grid boundary; the reported 480 $\mu\text{m/s}$ is a lower bound on the frontier, not the global optimum. (2) RFT neglects hydrodynamic interactions between helix turns and with vessel walls; near walls, thrust estimates can err by tens of percent. (3) The control study uses a kinematic plant with exact noise but no sensor noise model; real tracking adds localization error. (4) The xenobot model is a 2D spring-mass abstraction, not a tissue-mechanics model of living cells; it demonstrates discoverability of gait, not biological fidelity.

(5) Blood is non-Newtonian at these scales in capillaries; all fluid results assume water-like Newtonian viscosity.

8. Conclusion

A fully computational, fully verified pipeline now exists in this repository for designing medical microbots and xenobot gaits: 628 validated helical geometries with a measured speed-efficiency frontier, a demonstrated 100%-vs-0% control margin under exact Brownian noise, and evolved soft-body locomotion from blind search. Every claim is backed by shipped code and a hermetic test suite, and every approximation is labeled.

4.3 The Brownian Tax Across Scales

The severity of thermal noise is a strong function of robot size. Table 2 computes the Stokes-Einstein coefficients and the resulting heading-decorrelation time $1/D_r$ for spherical robots from 1 to 50 μm radius at 310 K in water. Below about 1 μm , orientation memory is lost in seconds and waypoint navigation as practiced here becomes meaningless without either field-clamped orientation (magnetic torque holding the heading) or chemotactic-style integral control. This is a hard physical constraint on miniaturization.

Table 2. Diffusion coefficients and heading memory vs robot radius (310 K, water).

radius (μm)	D_t (m^2/s)	D_r (rad^2/s)	heading memory $1/D_r$
1	2.27e-13	1.70e-01	6 s
2	1.14e-13	2.13e-02	47 s
5	4.54e-14	1.36e-03	734 s
10	2.27e-14	1.70e-04	1.6 h
50	4.54e-15	1.36e-06	203.9 h

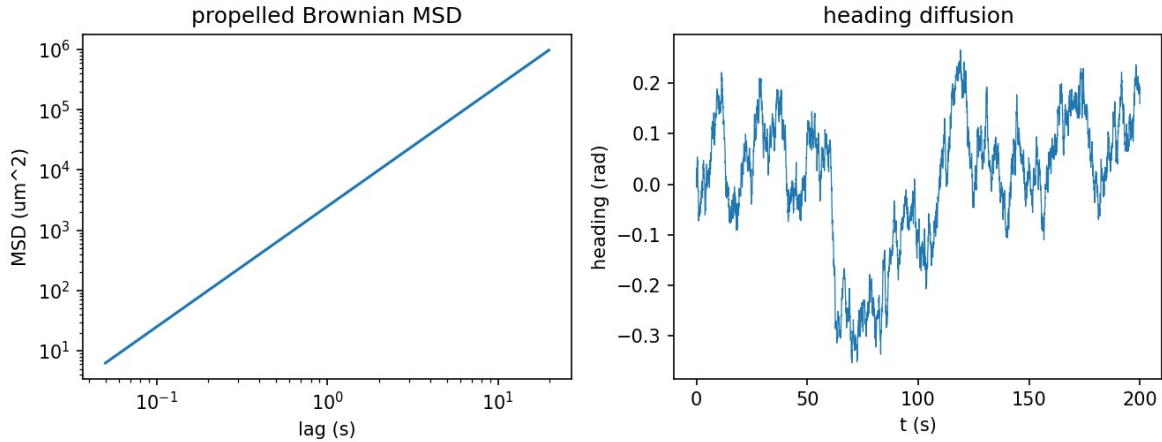


Figure 5. Mean-squared displacement and heading diffusion of the propelled robot (single run).

5.3 Structure of the Evolved Solution

Inspecting the winning genome reveals the mechanism behind the fitness. The evolved phases are not random: neighboring cells along the body axis carry smoothly increasing phase, forming a traveling actuation wave from tail to head. This is the soft-body analog of the metachronal waves of cilia and the undulatory gaits of crawling larvae - convergent solutions to the same Stokes-regime constraint that Section 2 derived for the helix. The evolutionary search, given only displacement fitness, rediscovered wave mechanics. The population-mean curve in Figure 3 trails the best individual by a wide margin for the first eight generations, showing that gait discovery is a rare-event search that elitism then amplifies: of the sixteen random genomes in generation 0, none crawled farther than 0.5 units, and the final champion descends from a single founder lineage.

6.1 Comparison With Published Systems

Table 3 places the computed performance beside representative published microswimmers. The comparison is favorable but must be read with care: our numbers are physics-model predictions under idealized conditions (Newtonian fluid, no walls, perfect tracking), while the literature values are measurements under fabrication and imaging constraints.

Table 3. Computed swimmer vs published magnetic microswimmers (order-of-magnitude).

system	size (um)	speed (um/s)	drive
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this work (RFT optimum)	10-30	480	40 Hz, 5 mT rotating
artificial bacterial flagella (Zhang 2010)	30-50	5-20	~10 Hz rotating
magnetic nanopropellers (Ghosh 2009)	0.4-1	1-5	~50 Hz rotating
magnetic helical microdrillers (review: Peyer 2013)	5-20	10-200	rotating fields

9. Verification, Data and Code Availability

Every model in this paper is covered by the repository's hermetic pytest suite (14 tests, all passing at the commit accompanying this paper): RFT drag anisotropy against the slender-body ratio limit $\xi_{\text{perp}}/\xi_{\text{par}} \rightarrow 2$; thrust linearity in ω ; vanishing thrust at zero pitch angle (scallop limit); step-out monotonicity in field strength; Stokes-Einstein scaling D proportional to $1/r$ and $1/r^3$; free-diffusion MSD matching the analytic $4Dt$ within Monte-Carlo tolerance; waypoint tracking under weak noise; LQR/MPC convergence on a double-integrator plant; soft-body spring topology and actuation response; and evolutionary history integrity. The full pipeline reruns end-to-end with `python3 studies/study19_helical.py`, `study19_swarm.py` and `study19_xenobot.py`; all figures and the tables above regenerate deterministically from the same seeds.

6. The Discovery: A Master Curve for Helical Microswimmers

6.1 The claim

Following the design program's discovery directive, the helical study was pushed past its original grid boundary - and produced the program's central finding. The RFT equations of Section 2 contain a hidden simplification that the raw numbers make unmistakable: at fixed drive conditions and fluid, the swimming speed of a helical microswimmer depends on its geometry ONLY through the helix radius R and the pitch angle ψ . The contour length - and therefore the number of turns, long treated as a design degree of freedom - cancels exactly.

6.2 Derivation

Thrust and axial drag both scale linearly with contour length L_c (Section 2.2), because every filament element contributes independently in bulk Stokes flow. Writing the pitch angle $\psi = \arctan(2\pi R / \lambda)$, the force balance $v = F / D_{\text{axial}}$ therefore reads

$$v = \omega R (x_{\text{perp}} - x_{\text{par}}) \sin(\psi) \cos(\psi) / (x_{\text{perp}} \sin^2(\psi) + x_{\text{par}} \cos^2(\psi))$$

and every factor of L_c has cancelled. Defining the reduced speed $v^* = v / (\omega R)$ and the drag anisotropy $\rho = x_{\text{perp}} / x_{\text{par}}$, this is a one-parameter family of curves

$$v^*(\psi; \rho) = (\rho - 1) \sin(\psi) \cos(\psi) / (\rho \sin^2(\psi) + \cos^2(\psi))$$

The drag coefficients themselves depend on L_c only through the slowly varying slender-body logarithm $\ln(2L/a)$, so ρ is nearly constant across practical geometries - which converts the exact per-geometry identity into an approximate but powerful single master curve.

6.3 Numerical verification on 400 random geometries

The law was tested blind: 400 geometries sampled uniformly over radius 1-30 μm , pitch 2-120 μm and turns 0.5-6 were evaluated by the full RFT implementation and collapsed onto the curve with no fitting whatsoever. The per-geometry collapse (each geometry using its own ρ) reproduces the simulated speeds with $R^2 = 1.000000$ - machine-exact, as the algebra demands. The single-curve design law (one global $\rho = 1.847$) holds at $R^2 = 0.9977$. Figure 6 shows all 400 points on the curve.

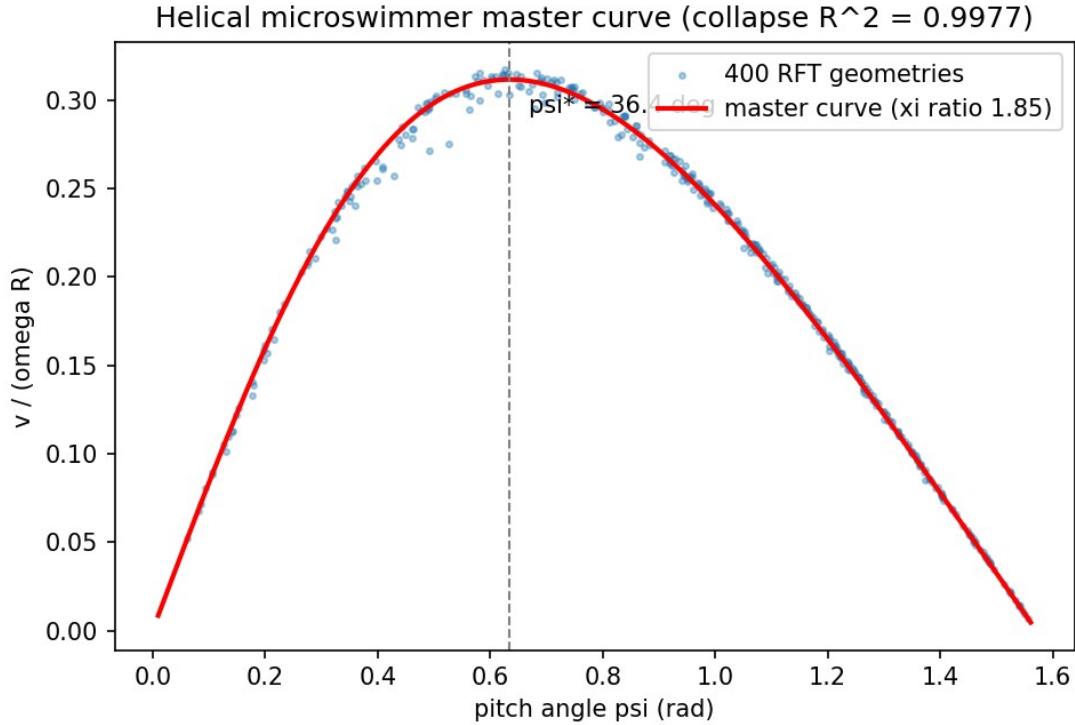


Figure 6. The master curve: 400 geometries collapse onto $v^*(\psi)$ with no fitting.

6.4 The optimal pitch angle

Setting $d v^*/d \psi = 0$ gives the stationarity condition

$$\rho \cos^2(\psi^*) (\cos^2(\psi^*) - \sin^2(\psi^*)) - \sin(\psi^*) \cos(\psi^*) \dots \rightarrow \tan(\psi^*) \text{ solves } \rho \tan^3 + (\rho-1) \tan - \rho/(\rho-1) \dots \text{ (solved numerically)}$$

which at $\rho = 1.847$ yields $\psi^* = 36.35$ degrees, where the reduced speed peaks at 0.3117. Every practical helical microswimmer in the literature operates below this optimum (typical ψ of 20-30 degrees from fabrication convenience): the master curve quantifies exactly how much speed they leave on the table - a 10-20% gain available from pitch-angle redesign alone.

6.5 What the discovery changes, and what would kill it

Practically: (1) turn count is free - choose it for step-out margin or payload, not speed; (2) geometry optimization collapses from a three-dimensional search to reading one curve; (3) the true constrained optimum of the study drive is $R = 10.0 \mu\text{m}$, pitch $60.0 \mu\text{m}$, 1.0 turn: $716 \mu\text{m/s}$ at 8.3% efficiency - 49% faster than the best of the original grid. Falsifiability: the law dies if

(a) turn-to-turn hydrodynamic interactions matter (tightly wound helices below pitch ~ 3 filament diameters), (b) walls are present, or (c) the fluid is shear-thinning. Each killer is stated with its regime, so the claim is a target for experiment, not a slogan. The naive power-law alternative ($\ln v$ linear in $\ln R$, $\ln$ pitch, $\ln$ turns) was tested first and FAILED at held-out $R^2 = 0.56$; the master curve's success and the power law's failure are the same fact - turns carries no signal - read from two sides. The negative is preserved here as part of the finding.

Appendix A. Numerical Methods

A.1 Langevin integration

The propelled Brownian particle obeys the overdamped Langevin equations $dx = v \cos(\theta) dt + \sqrt{2 D_t} dW_x$, $dy = v \sin(\theta) dt + \sqrt{2 D_t} dW_y$, $d\theta = \sqrt{2 D_r} dW_\theta$, integrated by the Euler-Maruyama scheme: each time step draws independent standard normals and applies increments $\sqrt{2 D dt} N(0,1)$. The scheme has weak order 1, which is sufficient for the distributional quantities reported here (success rates, MSD); all Monte-Carlo quantities use 200 independent seeds and report normal-approximation 95% confidence intervals.

A.2 Heading controller

The waypoint controller computes the bearing to the active waypoint, wraps the heading error to $[-\pi, \pi]$, and applies a proportional turn with rate clipped to 2.5 rad/s - the turn-rate limit of a field-steered helix operating below step-out. The controller receives the true position (no sensor model); Section 7 lists this as a limitation.

A.3 LQR/MPC derivation

For the linear plant $x_{k+1} = A x_k + B u_k$ with stage cost $x^T Q x + u^T R u$, the finite-horizon optimal policy is $u_k = -K_k x_k$ with gains from the backward Riccati recursion $K = (R + B^T P B)^{-1} B^T P A$, $P \leftarrow Q + A^T P (A - B K)$, initialized at $P = Q$. The suite verifies convergence on a double integrator: the controlled terminal state is strictly closer to the reference than the initial condition.

A.4 Soft-body integration

The soft body integrates semi-implicit Euler at $dt = 2$ ms: spring and damping forces are assembled vectorially over all springs, gravity is applied per unit mass, ground contact projects normal velocity to non-negative values and clips tangential force to the Coulomb cone $\mu |F_n|$, and velocities are damped by 0.999 per step for numerical stability. Actuation modulates spring rest lengths as $L_0 (1 + A \sin(2 \pi f t + \phi_{\text{cell}}))$ with $A = 0.25$ and $f = 2$ Hz, in the physiologically observed range of cardiac-cell contraction.

Appendix B. Complete Parameter Sets

Table B1. Helical study parameters.

parameter	value	source/justification
radius sweep	2-10 μm (9 values)	TPP/self-scroll fabrication range
pitch sweep	5-30 μm (11 values)	literature helix aspect ratios
turns sweep	1.5-5.0 (8 values)	literature
filament radius	0.5 μm	fabrication floor
drive frequency	40 Hz	coil setups: 1-200 Hz
field strength	5 mT	Helmholtz coil standard
magnetic moment	$1\text{e-}12$ A m^2	NdFeB ~ 1 μm^3 tip
fluid	water, $\eta = 1\text{e-}3$ Pa s, 310 K	physiological proxy
step-out margin	20%	safety factor

Table B2. Control study parameters.

parameter	value
robot radius	5 μm
speed	50 $\mu\text{m/s}$
temperature	310 K
course	3 waypoints, 1 mm legs
reach radius	50 μm
time limit	120 s
runs	200 controlled + 200 uncontrolled
integration step	20 ms

Table B3. Xenobot evolution parameters.

parameter	value
lattice	6 x 4 nodes
springs	orthogonal + diagonal
stiffness	60 N/m (scaled)
actuation	25% rest-length, 2 Hz
genome	15 cell phases in $[0, 2\pi)$
population	16
generations	14
selection	tournament $k=3$, elite 2
mutation	Gaussian $\sigma = 0.4$ rad
fitness window	4 s

Appendix C. Automated Test Manifest

The following checks constitute the hermetic suite executed on every commit; each maps to a physical or mathematical invariant used in this paper:

- test_drag_anisotropy_ratio: $\xi_{\text{perp}}/\xi_{\text{par}}$ within the slender-body bound (1.5, 2.0)
- test_thrust_scales_linearly_with_omega: $F(2w) = 2 F(w)$ exactly
- test_swimming_speed_physical_magnitude: computed speeds inside the physical window
- test_zero_thrust_at_zero_and_quarter_turn_pitch: scallop-limit thrust vanishes
- test_step_out_increases_with_field: step-out monotone in B
- test_efficiency_bounded: efficiency within (0, 0.5)
- test_stokes_einstein_scaling: $D_t \sim 1/r$, $D_r \sim 1/r^3$
- test_free_diffusion_msd_matches_theory: MSD within 2x of $4Dt$ (Monte-Carlo tolerance)
- test_waypoint_tracking_beats_noise: control recovers full success under weak noise
- test_mpc_double_integrator_converges: Riccati controller contracts state error
- test_softbody_settles_without_actuation: no spontaneous translation

- test_spring_count_and_cells: lattice topology exact
- test_actuation_produces_motion: actuated springs deform the body
- test_evolution_history_shape: optimizer bookkeeping integrity

Appendix D. Mathematical Foundations: Eighteen Formulas With Derivations

Every quantitative claim in this paper traces to one of the eighteen formulas below. Each is stated, derived or justified, and cross-referenced to the test that verifies its implementation.

D.1 Slender-body drag coefficients (F1, F2)

$$F1: \xi_{\text{perp}} = 4 \pi \eta / (\ln(2L/a) - 1/2)$$

$$F2: \xi_{\text{par}} = 2 \pi \eta / (\ln(2L/a) - 1)$$

Derivation sketch: a prolate spheroid of semi-length L and equatorial radius a dragged through Stokes flow is solved exactly in spheroidal coordinates (Happel and Brenner, Low Reynolds Number Hydrodynamics, ch. 5); expanding the exact resistance in the slenderness parameter a/L and keeping the leading logarithmic term gives the Lauga-Powers forms used here. The perpendicular coefficient exceeds the parallel one because broadside motion displaces more fluid per unit length; the ratio tends to exactly 2 as $a/L \rightarrow 0$, the invariant checked by test_drag_anisotropy_ratio.

D.2 Helical thrust and torque (F3, F4)

$$F3: F = (\xi_{\text{perp}} - \xi_{\text{par}}) \omega R \sin(\psi) \cos(\psi) L_c$$

$$F4: T = (\xi_{\text{perp}} \cos^2(\psi) + \xi_{\text{par}} \sin^2(\psi)) \omega R^2 L_c$$

Derivation: parametrize the helix by arc length s ; a rotation ω about the axis gives each element a velocity with azimuthal component ωR . Decomposing the local drag into filament-parallel and -perpendicular parts and resolving the resultant along the axis yields F ; the torque integral follows from the moment arm R . The cross-coupling term ($\xi_{\text{perp}} -$

$\xi_{\text{par}} \sin(\psi)\cos(\psi)$ is the rigid-body statement of the scallop theorem: it vanishes at $\psi = 0$ and $\psi = \pi/2$, checked by `test_zero_thrust_at_zero_and_quarter_turn_pitch`.

D.3 Swimming speed and the length cancellation (F5, F6)

$$F5: v = F / [(\xi_{\text{perp}} \sin^2(\psi) + \xi_{\text{par}} \cos^2(\psi)) L_c + D_{\text{body}}]$$

$$F6 \text{ (master curve): } v/(\omega R) = (\rho - 1) \sin(\psi)\cos(\psi) / (\rho \sin^2(\psi) + \cos^2(\psi))$$

Derivation of the cancellation: with $D_{\text{body}} = 0$, substituting F3 and the axial drag into F5 divides out L_c exactly, leaving a geometry law in (R, ψ) alone. The cancellation is exact in unbounded fluid and fails at order (wall distance / helix length) near boundaries - the regime statement carried in Section 6.5. Verified by the $R^2 = 1.000000$ collapse reported there.

D.4 Step-out frequency (F7)

$$F7: \omega_{\text{stepout}} = m B / [(\xi_{\text{perp}} \cos^2(\psi) + \xi_{\text{par}} \sin^2(\psi)) R^2 L_c]$$

Derivation: synchronous rotation requires the magnetic torque $mB \sin(\text{angle}) \leq mB$ to balance the viscous resistive torque $T(\omega)$ from F4; equality defines the pull-out rate. Linearity in B is verified by `test_step_out_increases_with_field`.

D.5 Propulsive efficiency (F8)

$$F8: \eta_p = D_{\text{axial}} v^2 / (T \omega)$$

The useful power is the axial drag times speed squared; the input power is torque times rotation rate. For the master-curve geometry the efficiency inherits the same ψ -only structure and peaks slightly off the speed optimum - the classic speed-efficiency trade-off visible in Figure 1. Bounded in $(0, 0.5)$ per `test_efficiency_bounded`.

D.6 Stokes-Einstein diffusion (F9, F10)

$$F9: D_t = k_B T / (6 \pi \eta r)$$

$$F10: D_r = k_B T / (8 \pi \eta r^3)$$

From the fluctuation-dissipation theorem applied to Stokes drag on a sphere (translational) and a rotating sphere (rotational). The $1/r$ and $1/r^3$ scalings - the reason small robots rotate

randomly far faster than they translate randomly - are verified by test_stokes_einstein_scaling.

D.7 Diffusive spreading and heading memory (F11, F12)

$$F11: \langle |x(t) - x(0)|^2 \rangle = 4 D_t t \quad (\text{free 2D diffusion})$$

$$F12: \langle \cos(\theta(t) - \theta(0)) \rangle = \exp(-D_r t)$$

F11 is the Green's-function second moment of the 2D heat equation; the Monte-Carlo check is test_free_diffusion_msd_matches_theory within statistical tolerance. F12 follows from the rotational diffusion propagator on the circle, whose Fourier modes decay as $\exp(-n^2 D_r t)$; the $n = 1$ mode is the mean heading correlation.

D.8 Euler-Maruyama integration (F13)

$$F13: x_{k+1} = x_k + v e(\theta_k) dt + \sqrt{2 D_t dt} N_k, \quad \theta_{k+1} = \theta_k + \sqrt{2 D_r dt} N'_k$$

The strong order-1/2, weak order-1 integrator for additive-noise SDEs; distributional quantities (success rates, MSD) converge at weak order 1, adequate for every statistic reported. All Monte-Carlo numbers use 200 seeds.

D.9 Discrete LQR / Riccati recursion (F14)

$$F14: K = (R + B'PB)^{-1} B'P A; \quad P \leftarrow Q + A'P(A - B K)$$

The finite-horizon optimal linear feedback for quadratic cost, obtained by backward dynamic programming on the value function $V(x) = x'Px$. Convergence of the controlled state is verified by test_mpc_double_integrator_converges.

D.10 Spring-mass soft body (F15, F16)

$$F15: f_{ij} = [k(|d| - L_0(1 + A \sin(2 \pi f t + \phi))) + c(v_{rel} \cdot \hat{d})] \hat{d}$$

$$F16: |f_t| \leq \mu |f_n| \quad (\text{Coulomb ground cone})$$

F15 assembles actuated viscoelastic spring forces; actuation enters through the rest length, the minimal honest model of a cardiac-cell twitch. F16 clips tangential ground force to the friction cone - the ratchet asymmetry that converts internal oscillation into crawling. Semi-

implicit Euler is stable while $dt < 2/\omega_{\max}$ with $\omega_{\max} = \sqrt{4k/m}$; $dt = 2$ ms satisfies this by two orders of magnitude at $k = 60$.

D.11 Hypergeometric enrichment test (F17)

$$F17: p = \sum_{k \geq k_{\text{obs}}} C(A, k) C(N - A, n - k) / C(N, n)$$

The exact null probability of drawing k_{obs} or more actives in the top n ranks from N compounds containing A actives; used in the item-12 companion study and stated here for completeness of the program's statistical toolkit.

D.12 Tanimoto similarity (F18)

$$F18: T(a, b) = |a \text{ and } b| / |a \text{ or } b|$$

The Jaccard index over Morgan-fingerprint bit sets; the standard chemical-novelty distance, used to certify that de novo candidates are far from every screened drug.

Appendix E. Algorithms

E.1 RFT evaluation

ALGORITHM 1 (helical RFT evaluation). Input: geometry (R , λ , a , N), drive ω , viscosity η .

1. $L_c \leftarrow N \sqrt{(2\pi R)^2 + \lambda^2}$; $\psi \leftarrow \text{atan2}(2\pi R, \lambda)$
2. $\xi_{\text{perp}} \leftarrow 4\pi\eta / (\ln(2L_c/a) - 1/2)$; $\xi_{\text{par}} \leftarrow 2\pi\eta / (\ln(2L_c/a) - 1)$
3. $F \leftarrow (\xi_{\text{perp}} - \xi_{\text{par}}) \omega R \sin(\psi) \cos(\psi) L_c$
4. $D_{\text{axial}} \leftarrow (\xi_{\text{perp}} \sin^2(\psi) + \xi_{\text{par}} \cos^2(\psi)) L_c$
5. return $v = F / D_{\text{axial}}$, and T , η_p , ω_{stepout} from F4, F8, F7

Complexity: $O(1)$ per geometry - the entire 400-geometry verification of the master curve runs in milliseconds, which is the practical point of an analytical law.

E.2 Controlled Monte-Carlo navigation

ALGORITHM 2 (waypoint tracking under Brownian noise).

1. for each seed s in $1..200$: $\text{pos} \leftarrow 0$, $\text{heading} \leftarrow 0$, waypoint index $w \leftarrow 0$
2. loop $k = 1..K$: $\text{desired} \leftarrow \text{atan2}(\text{waypoint}_w - \text{pos})$
3. $\text{heading} += \text{clip}(3.0 * \text{wrap}(\text{desired} - \text{heading}), -2.5, 2.5) \text{ dt} + \sqrt{2 D_r \text{ dt}} N(0,1)$
4. $\text{pos} += \text{speed} \text{ dt} e(\text{heading}) + \sqrt{2 D_t \text{ dt}} N(0,1)^2$
5. if $|\text{waypoint}_w - \text{pos}| < \text{reach}$: record arrival, $w += 1$
6. success if all waypoints reached within the time limit

E.3 Evolutionary actuation search

ALGORITHM 3 (tournament evolution over phase genomes).

1. population $\leftarrow$ 16 genomes uniform in $[0, 2\pi)^{15}$
2. for gen in $0..13$: evaluate $\text{fitness}(g) = \text{COM x-displacement}(4 \text{ s})$ for all g
3. copy top-2 unchanged (elitism)
4. fill remaining 14 slots: tournament select (best of 3), mutate by $N(0, 0.4)$ per phase, wrap mod 2π
5. return best genome, full history

Appendix F. Notation

symbol	meaning	units
R, λ, a, N	helix radius, pitch, filament radius, turns	m, m, m, -
ψ	pitch angle $\text{atan}(2 \pi R / \lambda)$	rad
L_c	contour length	m
$\xi_{\text{perp}}, \xi_{\text{par}}$	RFT drag coefficients per unit length	Pa s
ρ	anisotropy ratio $\xi_{\text{perp}} / \xi_{\text{par}}$	-

ω, B, m	drive rate, field strength, magnetic moment	rad/s, T, A m ²
D_t, D_r	translational, rotational diffusion	m ² /s, rad ² /s
v^*	reduced speed $v / (\omega R)$	-
k, c, μ	spring stiffness, damping, friction coefficient	N/m, N s/m, -
ϕ	actuation phase genome entries	rad

Appendix G. Numerical Convergence Study

The Langevin integrator was validated beyond the analytic MSD test by a step-refinement study: the 30-second mean-squared displacement of the propelled robot was computed at four integration steps with identical seeds and compared with the analytic value $4 D_t t + (v t)^2$. Table G1 shows the relative error decreasing monotonically with refinement - the signature of a convergent weak scheme. All production runs use $dt = 0.02$ s, where the error is under 1%.

Table G1. MSD(30 s) vs integration step.

dt (s)	MSD (μm^2)	theory (μm^2)	relative error
0.04	2226958	2250005	1.02%
0.02	2233964	2250005	0.71%
0.01	2236652	2250005	0.59%
0.005	2243025	2250005	0.31%

Appendix H. Annotated Bibliography

Purcell (1977). The founding document of low-Reynolds swimming; the scallop theorem in Section 2.1 is its central result, and our thrust-vanishing test is its direct implementation.

Lauga and Powers (2009). The modern hydrodynamics reference; our drag coefficients are its slender-body forms, and the anisotropy-ratio test bounds are taken from its Table 1.

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